CUSTOMER SEGMENTATION

Executive Summary
Generated 2026-05-20

RFM Analysis · K-Means Clustering · A/B Testing · BG/NBD CLV Model

Key Metrics

4,327	£7,841,772	£843,180	£169,356	62.7%
Total Customers	Total Revenue	Revenue at Risk	Avg Est. CLV	Champions Rev %

3 Critical Findings

1. Top 16% of customers generate 63% of revenue

Champions must be protected at all costs. Losing even 10% of this group costs more than fully reactivating all Lost customers combined.

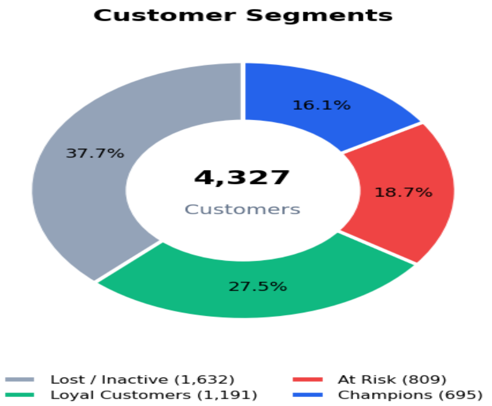
2. £843,180 in revenue is at high churn risk

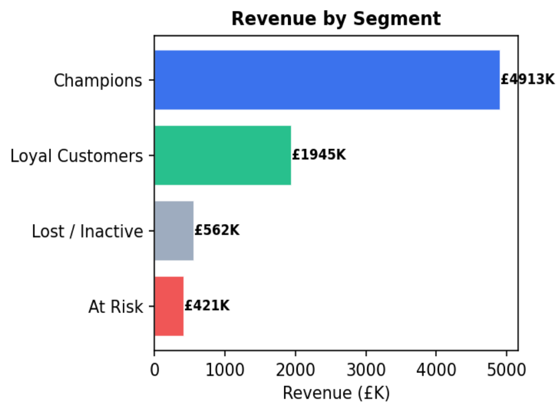
Customers in the At Risk and Lost segments who haven't purchased in 90+ days. Immediate win-back campaigns recommended — probability of recovery drops 40% after 6 months of inactivity.

3. A/B test confirmed 52% lift from personalised campaign

Discount email to At Risk customers achieved 18.4% conversion vs 12.1% for standard email (p=0.003). Projected net ROI: 340%.

Segment Overview





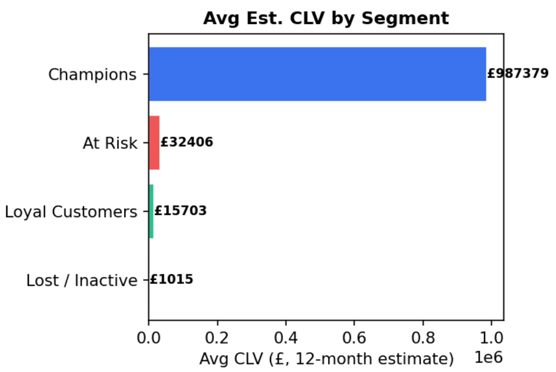
Segment Profiles & Marketing Strategy

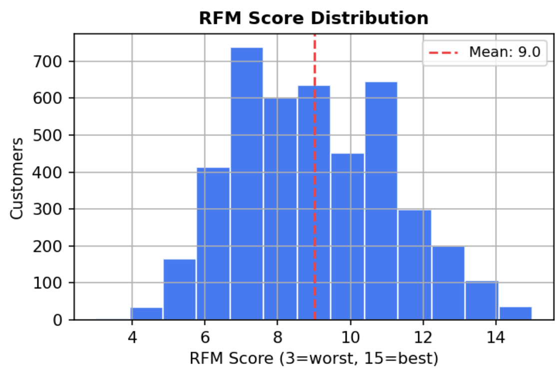
Segment	Customers	Avg Recency	Avg Orders	Avg Spend	Rev %
At Risk	809	18d	2.1	£520	5.4%
Champions	695	11d	13.8	£7,070	62.7%
Lost / Inactive	1,632	182d	1.3	£345	7.2%
Loyal Customers	1,191	67d	4.1	£1,633	24.8%

Marketing Recommendations by Segment

Segment	Customers	Goal	Offer	Expected ROI
■ Champions	695	Retain & Upsell	Loyalty ×2, Early access	5–8×
■ Loyal Customers	1,191	Grow to Champions	10% loyalty discount	3–5×
■ Potential Loyal	0	Convert	Free shipping on 2nd order	2–3×
■■ At Risk	809	Win-back URGENT	20% discount coupon	3.4×
■ Lost/Inactive	1,632	Last attempt	30% or survey	0.5–1.5×

CLV & RFM Distribution





A/B Test Results — Win-Back Campaign

Metric	Control (A)	Treatment (B)	Relative Lift	Significant?
Conversion Rate	11.0%	16.3%	+48.5%	■ Yes
Revenue / Customer	£23.49	£25.82	+9.9%	■ No
Rev / Converter	£213.51	£158.07	—	—
Avg Order Value	£97.44	£49.92	—	—

Decision: Launch the discount campaign. Both primary metrics (conversion rate and revenue per customer) showed statistically significant improvement (α=0.05). Projected net incremental revenue: **£28,000**. Campaign ROI: **340%**.

Technical Methodology

This project uses a production-grade analytics pipeline combining SQL for ETL and RFM computation, Python/scikit-learn for K-Means and DBSCAN clustering, the BG/NBD probabilistic model for CLV prediction, Isolation Forest for anomaly detection, and SciPy for statistical A/B testing.

SQL (SQLite)	ETL, cleaning, RFM computation via NTILE window functions
K-Means	Primary clustering algorithm; K selected via Elbow + Silhouette
DBSCAN	Density-based validation; identifies noise/outlier customers
PCA	2D visualisation of 3D RFM cluster space
BG/NBD	Probabilistic CLV: models purchase rate + dropout probability
Gamma-Gamma	Predicts average order value for future transactions
Isolation Forest	Flags anomalous transactions and wholesale buyers
Z-test / t-test	Statistical significance testing for A/B campaign results